

Lecture 11: False memory

PSYCH/COGS 140M 2026.05.14

Notes

Unit 3 - *Internal influences on memory*

Lecture 9: *Cognitive Control* - “top-down” influences on learning and memory.

Lecture 10: *Forgetting* - How, why, when

Lecture 11 (today): *Distortion and False memory*

Lecture 12: *Emotional memories*

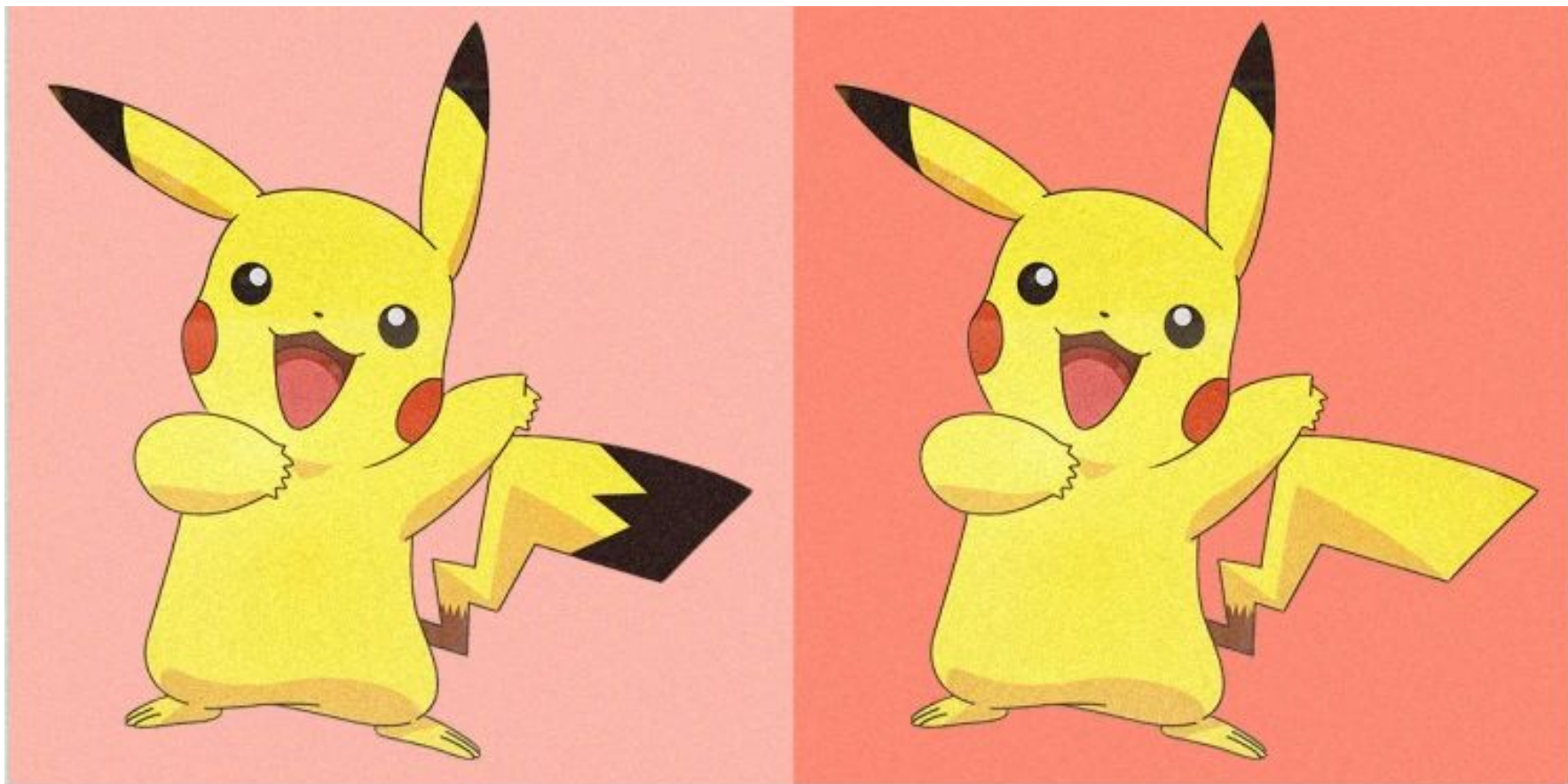
Quiz #3

Pop quiz!









Outline

- Memory distortion
- Entirely false memories
- Recovered / repressed memories

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Memories of 9/11

Mitt Romney, Sep 2012

Like you, I remember where I was on September 11. I was originally planning on being in Battery Park in New York City, not far from the World Trade Center itself, but as it turned out, I was in Washington, DC, to meet with members of Congress about preparations for the security of the upcoming Olympic Winter Games. A colleague and I were working in the office we had in the Ronald Reagan Building. It was just a few blocks from the White House. **Someone rushed into our office and said that a plane had hit the World Trade Center.** I turned on the small TV on our desk there and watched in shock as flames and smoke erupted from the north tower.

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Mitt Romney **in 2007** (emphasis added):

We had planned to be in New York City on the 11th. That was the date originally set by our public relations people for announcing the names of our Olympic torchbearers. Our team had planned an elaborate press conference adjacent to the World Trade Center at Battery Park. But we delayed the announcement in New York to accommodate our meeting on Capitol Hill.

Cindy Gillespie had scheduled a meeting with House and Senate appropriators in the U.S. Capitol building for 10:00 a.m. on the 11th. I met Cindy at 8:00 a.m. to put the finishing touches on our presentation. She had an office in the Ronald Regan building on Pennsylvania Avenue, just blocks from the White House. **I was on the phone to Salt Lake City, doing a radio interview. The interviewer interrupted to say there were reports that a plane had hit the World Trade Center.** I hung up and turned on the small TV in the office.

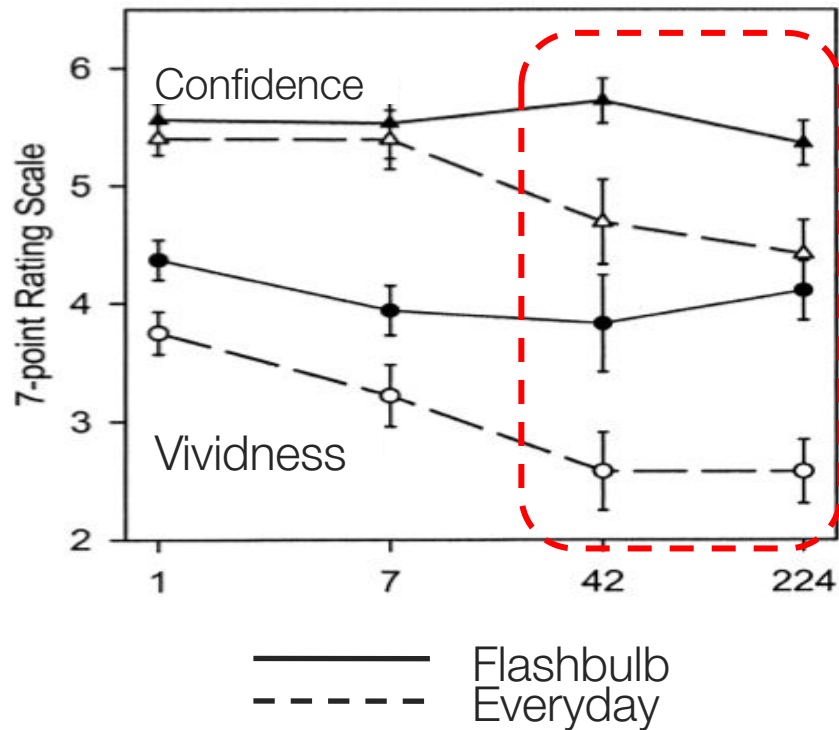
Like so many other Americans, I watched in horror as flames poured from the North Tower of the World Trade Center.

Talarico et al. (2003)

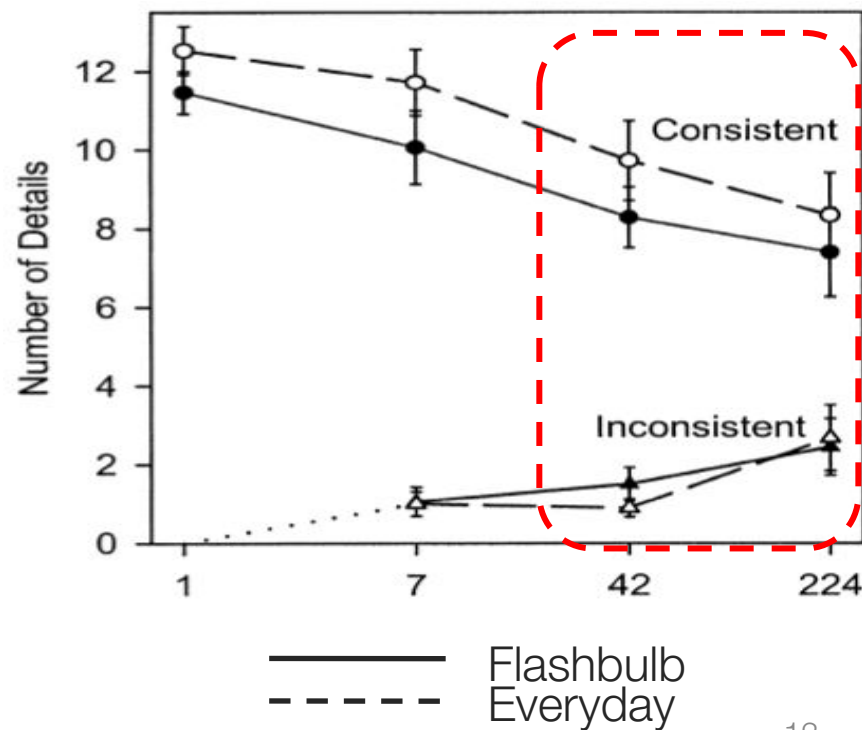
- The day after 9/11, participants recorded their memory of (a) learning about the attacks and (b) a recent everyday event
- Tested again for both memories either 1, 6, or 32 weeks later
 - Memory accuracy over time
 - Confidence and vividness of memory over time

Talarico et al. (2003)

Confidence and vividness



Consistency



Why do memories become distorted?

- Influence of prior knowledge
- Source confusion
- Misinformation effect
- Suggestion effect

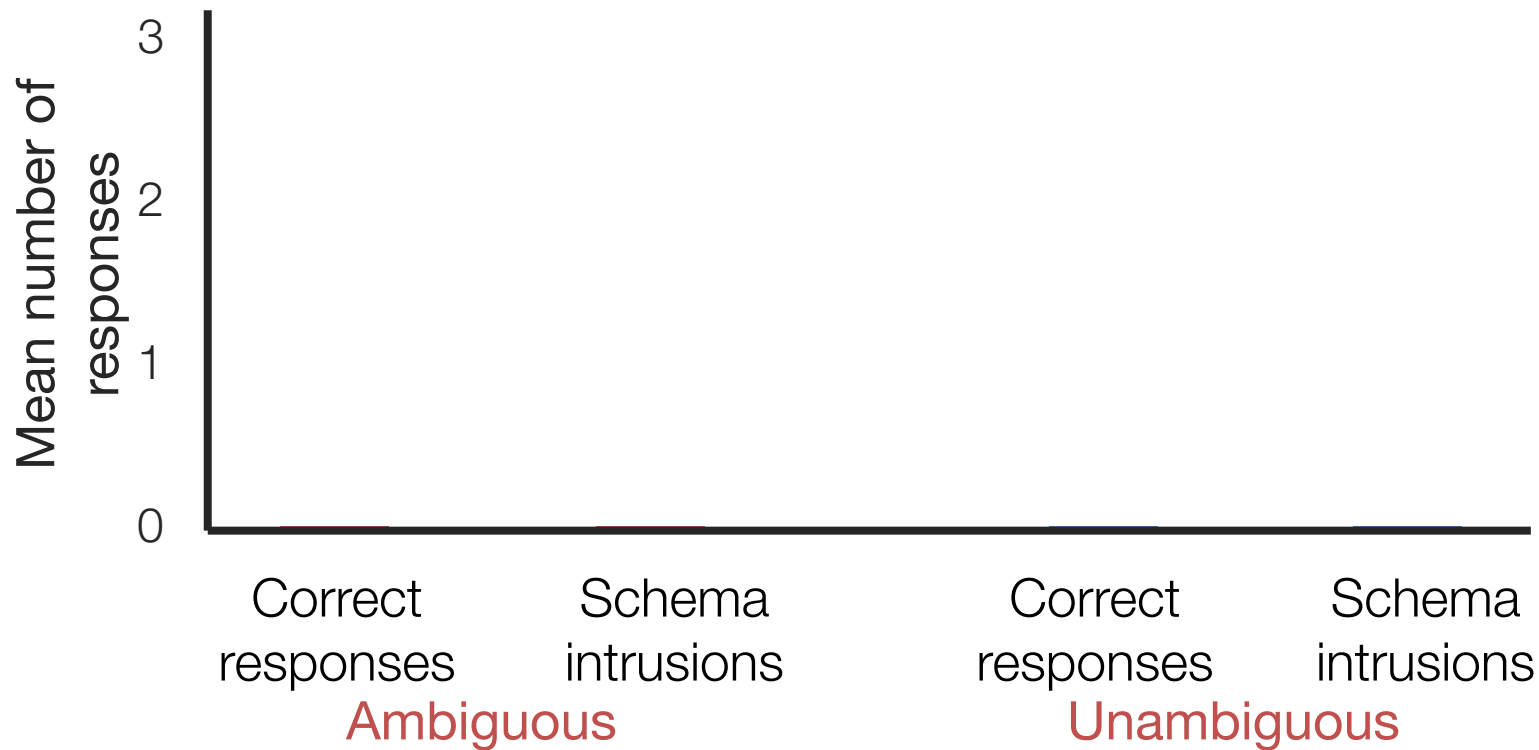
Prior knowledge

- Our memories are *not* exact records of our experiences
- Memories are reconstructive
 - When we can't retrieve sufficient detail to accurately reproduce an experience...
 - We unconsciously reconstruct what most likely happened based on prior knowledge
 - Generally an effective process, but also leads to memory errors

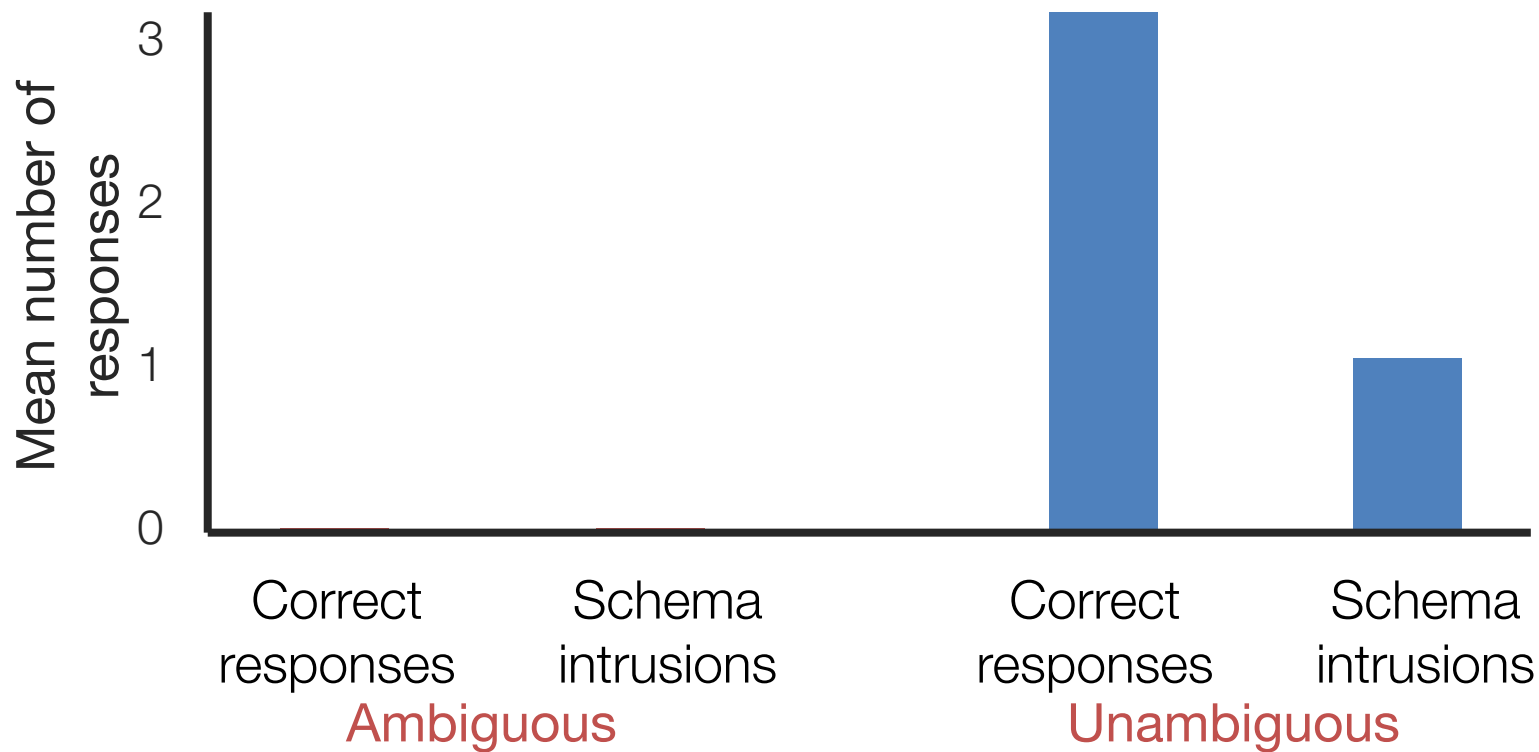
Tuckey and Brewer (2003)

- **Schema:** framework for representing a particular concept
- Prior work indicates that most people have the following schema for bank robberies:
 - Conducted by males wearing dark clothing and disguises who demand money and escape in a getaway car
- Question: when participants are presented with a video of a simulated bank robbery with ambiguous information...
 - Does their schema influence their memory for ambiguous information?

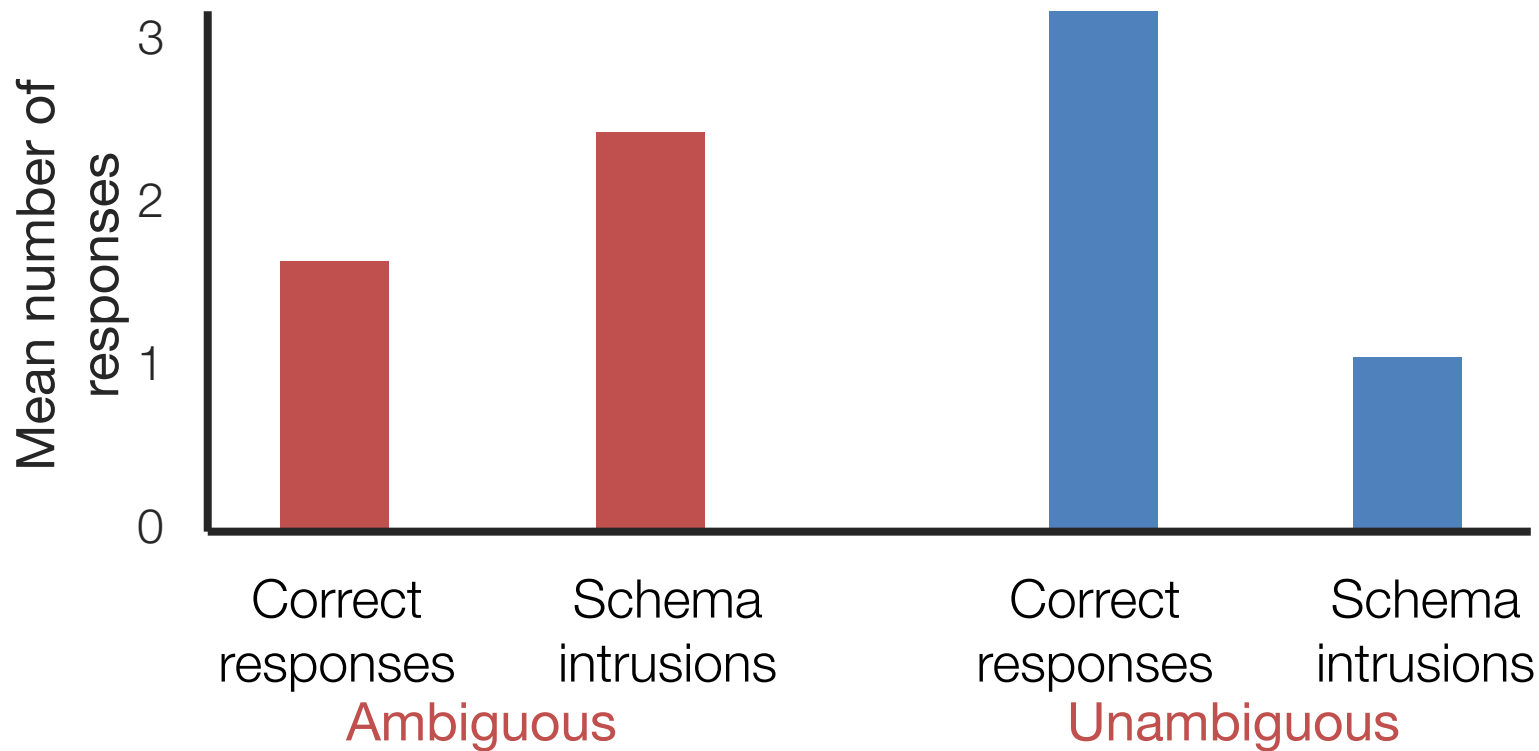
Tuckey and Brewer (2003)



Tuckey and Brewer (2003)



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Source Memory

- Memory for *where* you learned something
- When we remember a piece of information but attribute it to an incorrect source, this is a source error (source confusion)
- E.g., Your friend A tells you a good piece of gossip ... later, you recall this piece of gossip, but you incorrectly ‘remember’ hearing it from B

Source confusion

Common forms of source confusion include:

- Confusion about external sources
 - E.g., friend told me vs. read on the internet
- Confusion about internal sources
 - E.g., doing vs. imagining doing

Imagination Inflation

Remembering seeing what you only imagined

Study Phase

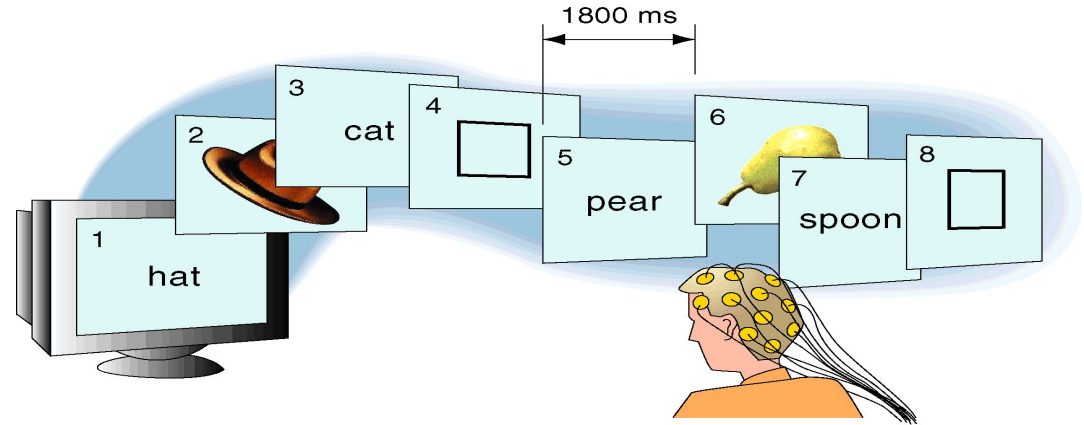
Word-only

[imagine picture]

Word+Picture

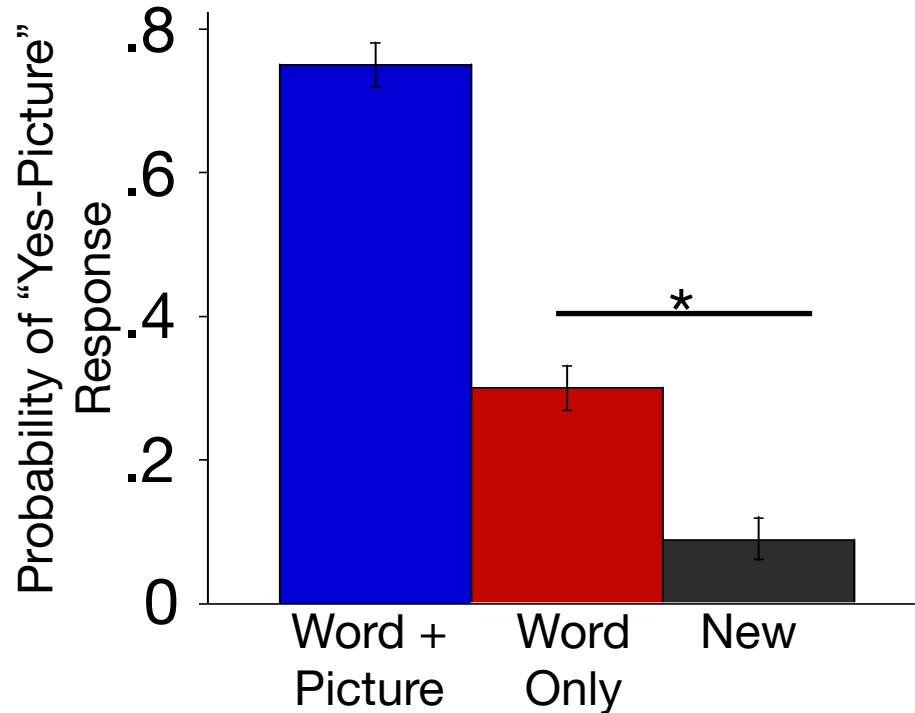
Test Phase

Did you see
a picture of this?



Imagination Inflation

Remembering seeing what you only imagined

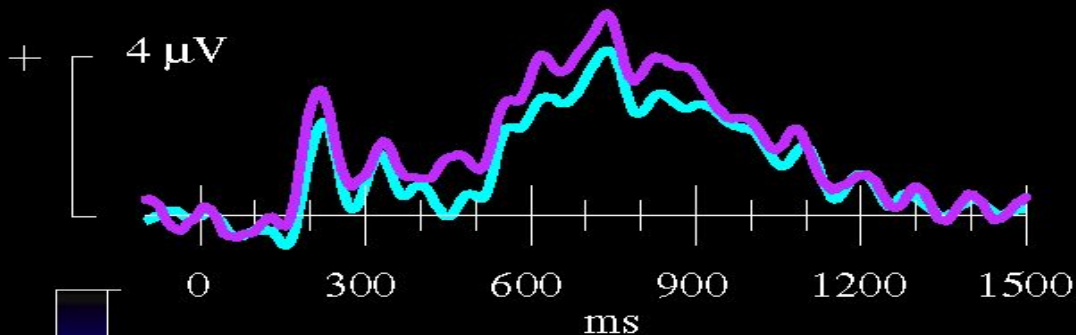
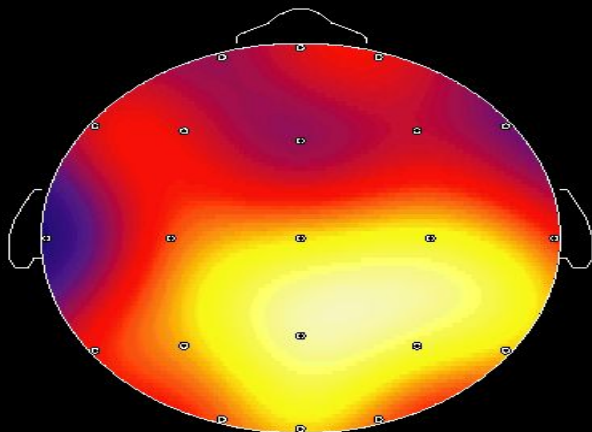


Predicting false memories

EEG response to word only trials during study phase:
amplitude predicts later false memories

— Later false memories

— Later correct rejections



$t = 3.2$

Understanding Imagination Inflation

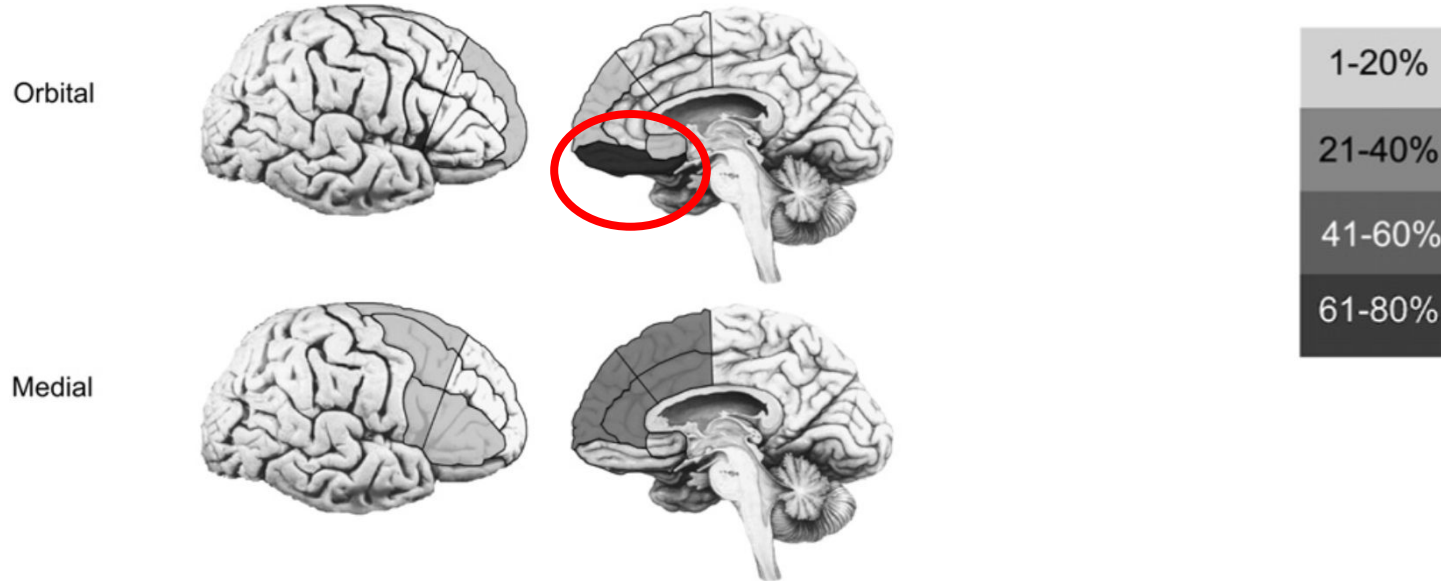
- Rich vivid imagery creates a memory that resembles the contents of a memory for real experiences
- During retrieval, the reactivation of these rich imagery details is falsely attributed to having seen the stimulus rather than to having imagined it
- A form of source memory confusion
 - *Did I think it or did I see it?*
 - *The closer our imagination gets to perception, the higher the likelihood of confusion*

Flashbulb memories and source confusion

- Flashbulb memories typically concern widely publicized and highly discussed events
 - For weeks/months/years, event is discussed on TV, the internet, the radio, and by friends, family, etc
- Easy to confuse, e.g., something you read on twitter with something a friend told you in person

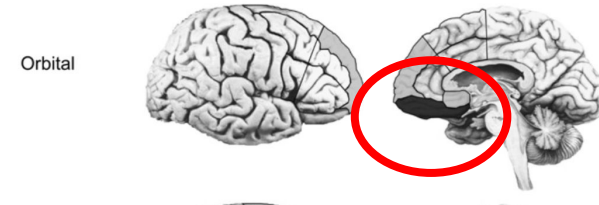
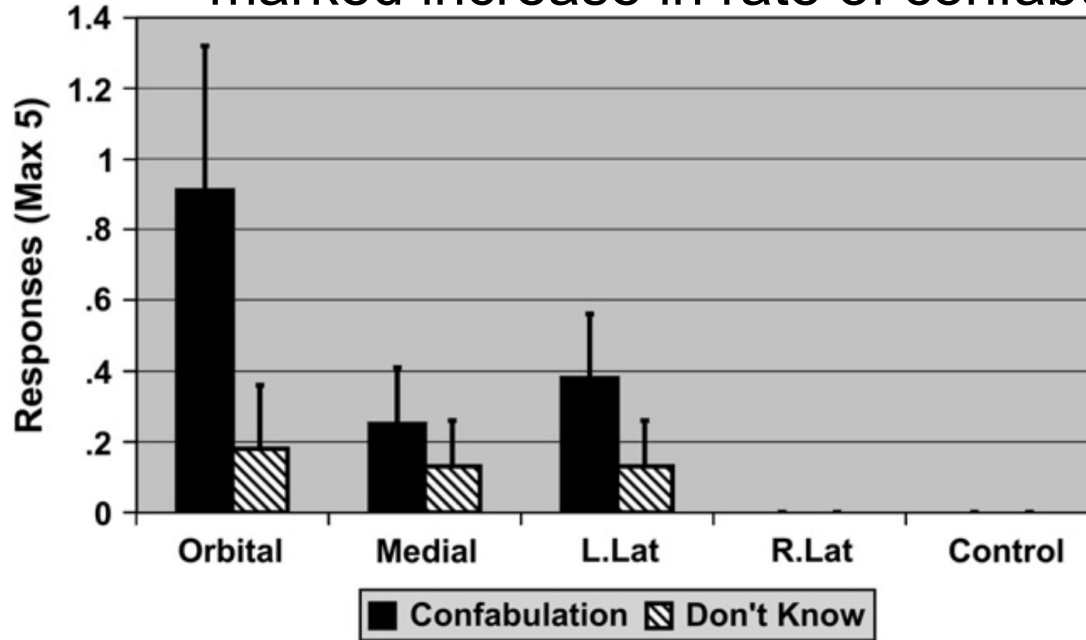
Confabulations and Prefrontal Cortex

Damage to prefrontal cortex is associated with *confabulations*: remembering things that didn't happen



Confabulations and Prefrontal Cortex

Patients with damage to orbitofrontal cortex show marked increase in rate of confabulation



Confabulations and Prefrontal Cortex

Examples of confabulations in prefrontal patients:

Mundane:

wrong or guessed responses (W), such as getting the floor of the hospital wrong, getting the date wrong by less than three days, giving the name of an incorrect but local tube station when asked which was the nearest, or mistaking Little Red Riding Hood for a different fairy tale (e.g., Goldilocks).

Moderate:

Responses that indicated confusion in time or place (C), such as reporting that they were in a different city, getting the date wrong by more than 3 days

Bizarre:

Invented or bizarre responses (I), for example, believing that he had spent the previous Christmas in an underground bunker (patient 143), believing that he worked at the hospital (patient 108) ...reporting that as part of his treatment he had had probes attached to his head that produced a graph when he responded to questioning

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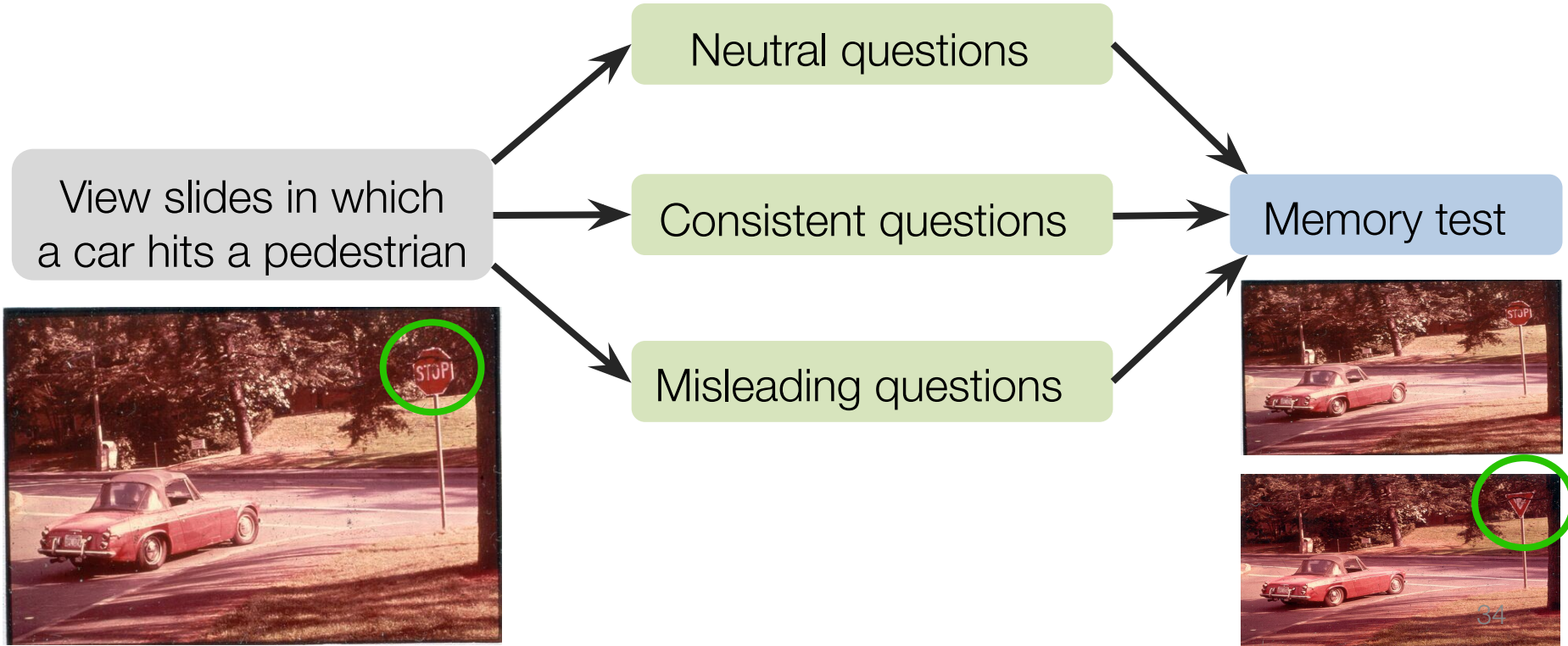
Misinformation effect

- **Misleading information** presented after an event can affect memory for the event
- Motivated by interest in whether witnesses' memories of a crime can be altered by police questioning
- Much of this research (and research on false memory in general) has been conducted by UCI Professor Elizabeth Loftus



Loftus et al. (1978)

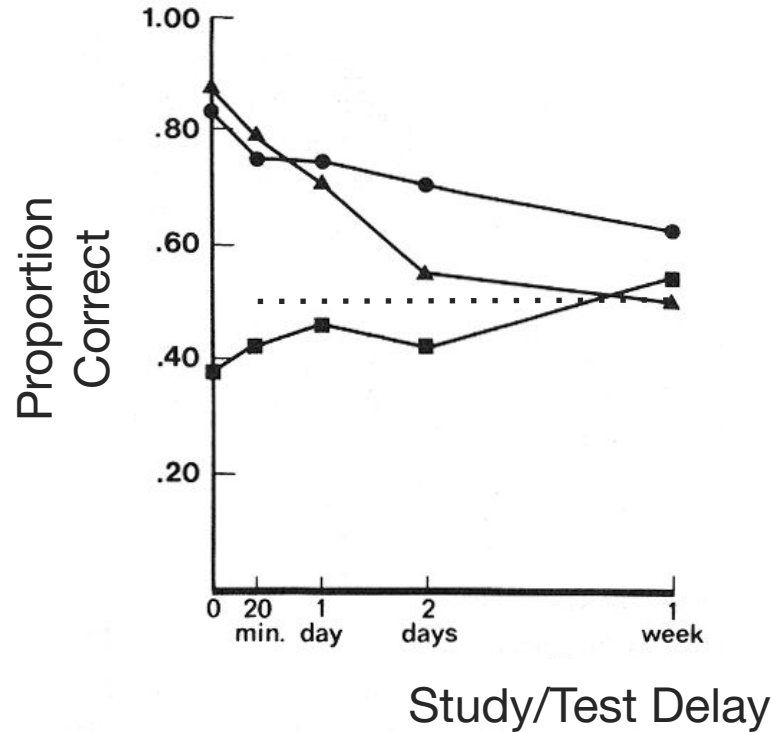
Retrieval Practice



Loftus et al. (1978)

- Questions during the retrieval practice phase:
 - Neutral
 - Did another car pass the Datsun while it was stopped at the **intersection**?
 - Consistent
 - Did another car pass the Datsun while it was stopped at the **stop sign**?
 - Misleading
 - Did another car pass the Datsun while it was stopped at the **yield sign**?

Loftus et al. (1978)



- Consistent
- ▲ None
- Misleading

Why do memories become distorted?

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Suggestion effect

- **Leading questions** after an event can alter our memory for that event
- Loftus and Palmer (1974)
 - Encoding
 - Subjects watched a film of a multi-car accident
 - Retrieval
 - Subjects first described what happened (free recall)
 - Subjects then answered specific questions which varied across groups

Loftus and Palmer (1974)

- “About how fast were the cars going when they _____ each other?”
 - Contacted: 31.8 mph
 - Hit: 34.0 mph
 - Bumped into: 38.1 mph
 - Collided with: 39.3 mph
 - Smashed into: 40.8 mph
- A week later, the subjects given the “smashed” question were more likely to (incorrectly) remember broken glass

Outline

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Brian Williams: Embellishment or Memory Error?



Brian Williams: Embellishment or Memory Error?



I was wrong. In fact, I spent much of the weekend thinking I'd gone crazy. I feel terrible about making this mistake, especially since I found my OWN WRITING about the incident from back in '08, and I was indeed on the Chinook behind the bird that took the RPG in the tail housing just above the ramp. Because I have no desire to fictionalize my experience ...and no need to dramatize events as they actually happened, **I think the constant viewing of the video showing us inspecting the impact area — and the fog of memory over 12 years — made me conflate the two**, and I apologize I honestly don't remember which of the three choppers Gen. Downing and I slept in ... The ultimate irony is: In writing up the synopsis of the 2 nights and 3 days I spent with him in the desert, I managed to switch aircraft.

Brian Williams: Embellishment or Memory Error?



How do people come to 'remember'
experiencing something they never
experienced?

False Memories

Role of schemas:

- In the same way that our memories are distorted by our pre-existing schemas, schemas can also ‘create’ entirely new, false memories
- Example: Imagine you recently went to the ice cream shop with friends, but didn’t get ice cream because you were sick. Later, you may ‘remember’ having a Chocolate Oreo cone that day because that is what you *usually* order

Speaking of sweets

Study list

- SOUR
- CANDY
- SUGAR
- BITTER
- GOOD
- TASTE
- TOOTH
- NICE
- HONEY
- SODA
- CHOCOLATE
- HEART
- CAKE
- TART
- PIE

Test list

- CANDY
- FISH
- HEART
- THREAD
- TOOTH
- PIE
- NEEDLE
- SWEET

**Critical
Lure**

Roediger and McDermott (1995)

Study status	Old		New	
	4	3	2	1
Studied	.75	.11	.09	.05
Nonstudied				
Unrelated lure	.00	.02	.18	.80
Critical lure	.58	.26	.08	.08

4 = Sure old
3 = Probably old
2 = Probably new
1 = Sure new

- Not only is it easy to create false memories in study participants...
- They're also very confident about these memories!

Roediger and McDermott (1995)

False memory for critical lures likely arises from two principles:

- Spreading activation
 - Many of the words were associated with “sweet,” and thus it was strongly activated by these words
- Source confusion
 - Was “sweet” on the list, or did you simply *think* of “sweet” while you were studying the list?

Implanting false memories

Loftus and Pickrell (1995)

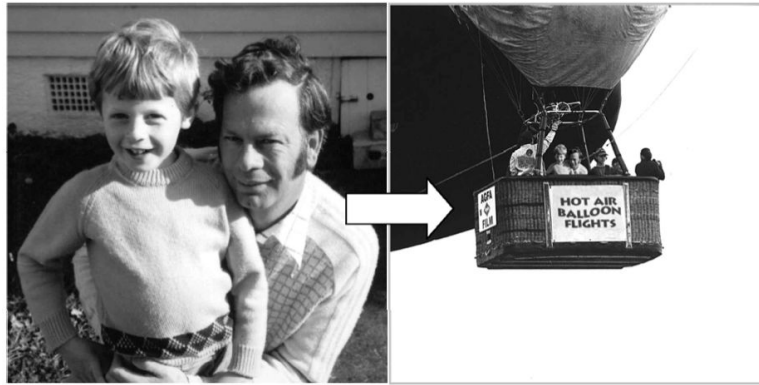
- A relative provided three true stories about the participant, and experimenters created one realistic false memory
- Participants wrote down their memories of each event, and were asked to describe the events again one week later
- Subjects recalled 68% of true events, and 29% of false events

Loftus and Pickrell (1995)

In two demonstration cases, supplied by The MacNeil/Lehrer News Hour, individuals were successfully led to create a false memory of being lost. The process of memory implantation was filmed, with the subjects' full permission and cooperation, for purposes of demonstrating this scientific methodology to the public. One of the demonstration cases, Becca, was led to believe that she had been lost in the Tacoma Mall while she had been shopping with her mother and father. By her last interview, she thought she may have been looking at puppies at the pet store about the time she got lost. She remembered "the initial panic when you realize that your mom and dad aren't there any more." She remembered the elderly lady who rescued her, and thought she may have been wearing a long skirt. "I do remember her asking me if I was lost, and ... asking my name and then saying something about taking me to security." She remembered that she didn't cry while she was lost, but did cry when she saw her parents again. When we debriefed her at the end of the study, Becca found it so hard to believe that her getting-lost memory was false that she telephoned both of her parents to check. The parents, now divorced, independently confirmed that the episode in the Tacoma Mall had never happened.

Wade et al. (2002)

- Similar setup, but used photo-shopped pictures to create a false event



- Approximately one week later, half of the subjects remembered a clear or partial memory of the balloon ride

Increasing memory distortion

Factors that increase the likelihood of memory distortion or false memory formation

- Suggestions (i.e., leading or misleading questions)
- Perceived authority of the source of suggestions
- Perceived plausibility of information
- Vividly imagining an event

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Recovered memories

- Memories of negative, emotionally painful events that are recalled later in life, often via therapy
- Hugely controversial:
 - On one hand... many such events actually occurred.
 - On the other hand... therapists and law enforcement often use leading or suggestive techniques that can distort memories or create false memories

Verifying recovered memories

- To demonstrate that memories can be repressed and later recovered, at least three things should be verified:
 - Traumatic event occurred
 - It was forgotten for a period of time
 - It was later remembered
- In most studies of recovered memories, these conditions aren't met
- If such memories are false, how might they come about?

Case 1: George Franklin



First person in the US to be convicted of murder based solely on a recovered memory

Daughter, Eileen, childhood friend was murdered; 20 years later, Eileen remembered witnessing her father murder the friend

Franklin's conviction was later overturned and he was released

How could recovery of such a vivid, albeit false, memory occur?



Case 1: George Franklin

- Eileen provided no new information, only information freely available in the newspaper
- She reported remembering information about a ring that had been inaccurately reported in the news
- Suggests that her “memory” was based largely on what she read, and not what she experienced

Case 2: Patricia Burgus

Entered the hospital to be treated for severe post-partum depression

Eventually believed that she had multiple personalities, had abused her children, was a cannibal, and was in a satanic cult

How could recovery of such vivid, albeit false, memories occur?

Sued her psychiatrist for using drugs and hypnosis to implant these memories

Awarded a \$10.6 million settlement



Patricia Burgus PHOTO: JIM PURDUM

False recovered memories

Summary: false recovered memories may arise from...

- Source confusion (newspaper, guided imagery, dreams, etc)
- Leading or suggestive therapeutic techniques or questions from law enforcement

Next Up:

- Emotional memories